Problem Chosen	2026 HiMCM/MidMCM Summary Sheet	Team Control Number
B		1111111

Measuring Useful Cooling in Open Systems

Working summary: Requirements 1 and 2

An open freezer case must maintain appropriate product temperatures despite heat exchange with its surroundings. Electricity use alone does not show how well it performs this task.

We distinguish thermal demand, electrical input, and delivered storage service. Infiltration, cabinet heat transfer, internal equipment, restocking, and defrost contribute to demand. Equipment condition modifies performance, while temperature differences across product locations determine whether the intended service is delivered.

We identify four quantities that define cooling service efficiency: energy input, service quantity, local service quality, and service duration. Compliant product mass-hours per kilowatt-hour provide one possible freezer measure. Product-specific temperature requirements remain a separate adequacy check. Comparisons require prescribed loading, matched ambient conditions, and complete operating cycles.

We extend this principle to a regular-size New York City bus shelter, using documented dimensions and a three-zone model. Fan-driven air movement and evaporative temperature/humidity changes feed into a common SET-based hot-discomfort proxy. Signed relief accumulated over passenger-hours is divided by electrical energy, with remaining discomfort reported separately.

The bus model is the fixed-background limit of a shared heat/moisture and local-delivery framework. Figures compare local conditions, electricity-normalized relief, humidity, passenger positions, delivery losses, and sensitivity. In the stated illustrative cases, the fan has the highest service efficiency, while the large evaporative cooler leaves the least residual discomfort. A delivery-loss stress test identifies conditions where the smaller evaporative cooler becomes more efficient. These are conditional scenario results, not measured device rankings. The power ratios give transparent break-even requirements: B and C must deliver respectively 4.8 and 10 times A's accumulated relief to exceed its service efficiency.

For Canberra MRT station, the same device, delivery and SET modules are retained while the background air and surfaces evolve with ventilation and passenger loads. Embedded local mixing regions distinguish delivered air from the surrounding background without counting heat or moisture twice. Device type, count and allocation are compared using electricity, remaining discomfort and coverage. Delivery and motor-heat sensitivities test how these assumptions affect selection; documented architecture is separated from assumed operating parameters.

Draft status. Requirements 1 and 2 are developed here. Requirements 3–5 and the final whole-paper summary remain to be completed. Numerical results are simulations under explicit assumptions; no field performance measurements are claimed.

Contents

1 Understanding Cooling Efficiency **3**
1.1 Selecting the quantities to evaluate . . . 3
1.2 Factors that explain performance . . . 4
1.3 Implications for the bus-stop model . . . 4
1.4 A common physical structure and a controlled extension . . . 5

2 Cooling a Bus Stop: From Local Conditions to Passenger Service **6**
2.1 Data, interpretation, and a reproducible scenario . . . 6
2.2 Predicting the local environment . . . 8
2.3 Converting local conditions to a common human response . . . 9
2.4 Measuring service and retaining a quality check . . . 10
2.5 Numerical results: local mechanisms and the efficiency tradeoff . . . 11
2.6 Climate and passenger distribution . . . 11
2.7 When the recommendation changes . . . 12
2.8 Sensitivity, verification, and scope . . . 13

3 Canberra MRT Station: A Semi-open Waiting-area Case **15**
3.1 Site evidence and the modeled waiting area . . . 15
3.2 What changes from the bus model, and why . . . 16
3.3 Coupled heat, moisture, and surface balances . . . 16
3.4 Local delivery without double-counting heat or water . . . 17
3.5 Equipment selection, allocation, and coverage . . . 18
3.6 Canberra scenario and comparable deployment tests . . . 18
3.7 Comparing service, coverage, and electricity . . . 19
3.8 Verification, evidence, and transfer . . . 21

References **24**

Report on Use of AI Tools **25**

1 Understanding Cooling Efficiency

An open freezer case succeeds by preserving its products. Low electricity use may reflect a smaller workload, shorter operation, or inadequate temperatures; high heat removal may reflect unwanted environmental heat gains. Cooling efficiency must therefore relate energy input to the storage service actually delivered.

Figure 1 makes the open-front geometry concrete: the front opening permits air exchange, the cabinet surfaces transmit heat, and the shelves separate product locations whose temperatures should be checked. The pictured model is a medium-temperature refrigerated display case [12]; its geometry illustrates these mechanisms, while freezer operating temperatures and product requirements must be specified separately.

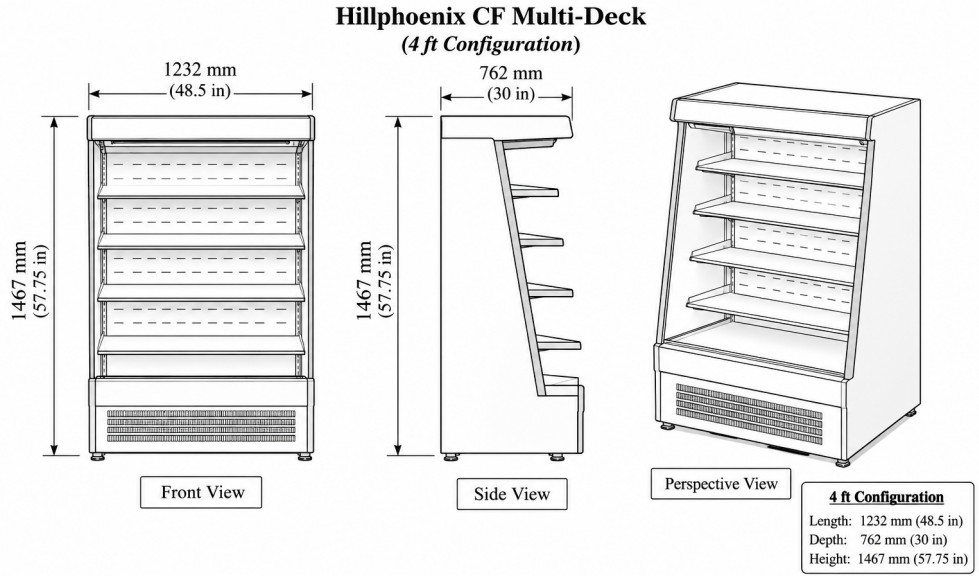


Figure 1: Open-front refrigerated display case: user-supplied schematic with external dimensions matching the Hillphoenix CF-R 4-ft configuration [12]. Millimetre labels are rounded; the internal shelf arrangement is schematic.

1.1 Selecting the quantities to evaluate

Table 1 identifies four quantities that enter the useful-service numerator or energy denominator in both settings.

Energy input includes auxiliaries and attributable plant electricity without overlapping meter readings. **Service quantity** accounts for workload, subject to prescribed loading limits.

Local service quality requires product temperatures at representative locations, rather than discharge-air temperature alone [1]. Averages can hide a warm shelf, and uniformity does not guarantee acceptable storage; product-temperature requirements remain essential [2].

Table 1: Evaluation quantities and their transfer to a bus stop.

Quantity	Open freezer case	Bus-stop shelter
Energy input	Electricity used by the case and its attributable refrigeration plant (kWh).	Electricity used by fans, pumps, and other cooling equipment (kWh).
Service quantity	Actual mass of products stored under a specified loading protocol (kg).	Number of people present in each occupied zone.
Local service quality	Product temperatures relative to product-specific storage requirements.	Thermal discomfort relative to the same setting without active cooling.
Service duration	Duration of product storage during the evaluation period (h).	Duration of passenger exposure while waiting (h).

Service duration distinguishes sustained performance from a favorable snapshot. Include restocking, compressor cycling, and defrost recovery, with comparable initial and final thermal states so that stored cold cannot inflate the result.

1.2 Factors that explain performance

The explanation follows the path from environmental loads to local conditions and then to delivered service. Its process-oriented organization draws on a supplied AI analysis, with the physical dependencies revised here [20].

From surroundings to local conditions. Infiltration carries sensible heat and moisture through the opening; walls transmit heat, and exposed surfaces exchange radiation. Geometry, insulation, ambient temperature, humidity, and airflow therefore alter the conditions around the products and the refrigeration demand [1].

From operation to local conditions. Lights and fans release heat, warmer incoming stock adds a temporary load, and defrost interrupts normal cooling [1]. Equipment condition changes insulation, airflow, and power demand. These mechanisms affect both local conditions and measured electricity; age itself is not an additional score.

From local conditions to service. Product placement and thermal response determine whether the food remains within its required range. Accumulating compliant product mass over time gives useful storage service. The same sequence applies to passengers, but their response depends on the thermal environment and their exposure, rather than food-temperature limits.

1.3 Implications for the bus-stop model

The preceding analysis motivates a common evaluation principle:

$$\boxed{\text{Cooling efficiency} = \frac{\text{useful service accumulated over objects and time}}{\text{energy input over the same period}}.} \tag{1}$$

For a prescribed product load, one concrete freezer measure is

$$\eta_f = \frac{\int \sum_i m_i(t)\mathbf{1}\{T_{p,i}(t) \in \mathcal{T}_i\}\, dt}{\int p_f(t)\, dt}, \tag{2}$$

where $\mathcal{T}_i$ is the product-specific admissible temperature range, t is in hours and p_f is in kW. Power includes attributable plant and auxiliary loads; comparable loading and complete operating cycles prevent a smaller workload or stored cold from inflating the score. Temperature excursions remain a separate adequacy check. Human service instead measures discomfort relief over passenger-hours. The two numerators describe different recipients and cannot be numerically ranked against one another.

For passengers, a fan changes air movement while evaporative equipment changes temperature and humidity [3, 4]. Weight the response by where people stand and how long they wait; empty space earns no service credit. Retain separate minimum-service checks, since partial relief can improve efficiency without providing adequate conditions. Compare strategies under matched weather, loading and evaluation periods. Water, cost and broader visitor experience enter the later effectiveness analysis.

1.4 A common physical structure and a controlled extension

All three settings exchange heat and moisture with their surroundings. For a fixed dry-air mass M_z and balanced incoming/outgoing air streams, a common control-volume structure is

$$M_z \dot{h}_z = \sum_j \dot{m}_{jz}(h_j - h_z) + S_{h,z}, \tag{3}$$

$$M_z \dot{w}_z = \sum_j \dot{m}_{jz}(w_j - w_z) + S_{w,z}. \tag{4}$$

Here h is J/kg dry air and w is kg vapor/kg dry air; S_h and S_w collect the consistently defined heat and water sources. Refrigeration, product heat transfer and possible condensation enter a freezer's sources. Fans, evaporation, people and surfaces enter the public-space sources. The recipient-response module changes from food-temperature compliance to human SET; conservation and service accounting remain the organizing principles. Requirement 1 establishes these boundaries rather than claiming a calibrated freezer simulation.

The bus model treats the outdoor air as a fixed reservoir and resolves steady local delivery. The rail model retains that local calculation but solves the background balances and surface storage over time. This follows the heat/moisture balance structure used in building simulation [13]. It also takes a process-level lesson from Cruz-Salas et al. [14]: distinguish evaporative treatment, air transport and the conditions in the occupied space. Their study concerns misting in a dry-season, unoccupied Mexican cafeteria; its water-loss factors and measured cooling are not parameters for our wet-pad devices or Singapore case.

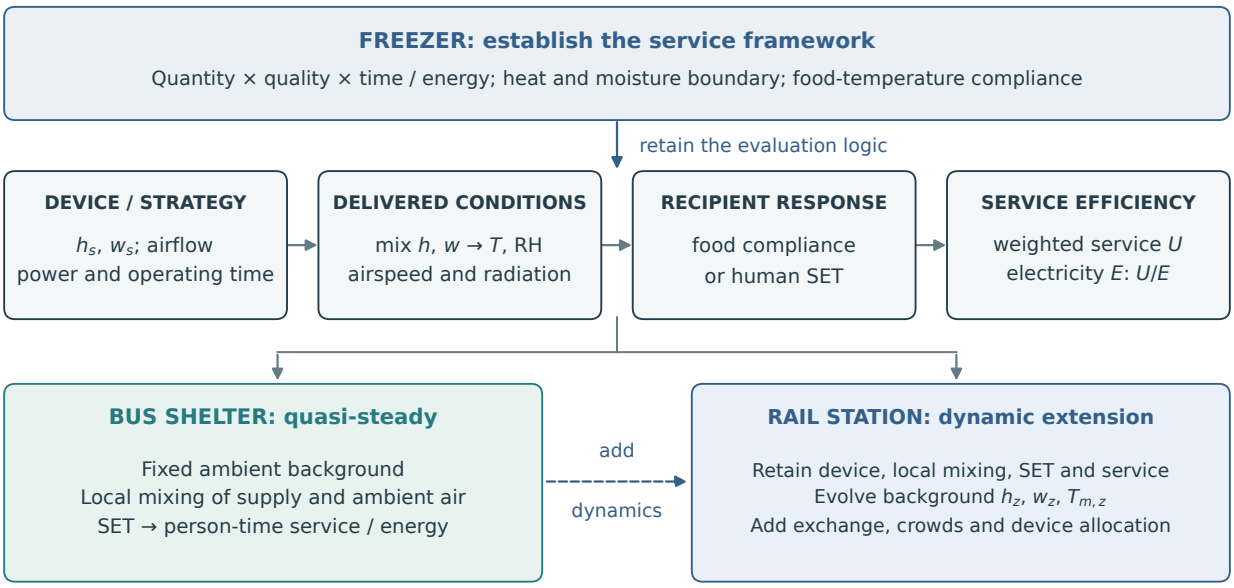


Figure 2: Model inheritance: common conservation and service accounting, object-specific responses, and a controlled extension from fixed to dynamic background air.

2 Cooling a Bus Stop: From Local Conditions to Passenger Service

We apply the service principle in Section 1 to Requirement 2(a). A device first changes the environment at each passenger's position; those changes alter thermal discomfort; the resulting relief is accumulated over people and time. This separates three questions: how much relief is delivered, where it is delivered, and how much electricity it requires.

We adopt the regular-size New York City shelter in Figure 3 as the common geometry for all three devices. Its documented dimensions provide a reproducible physical scale for the occupied zones.

The official dimensions are 14 ft by 5 ft by 8 ft 11 in [11]. Converting the imperial values before rounding gives $L = 4.2672$, $W = 1.5240$, and $H = 2.7178$ m. We use the gross plan area $A = LW = 6.5032\,\mathrm{m}^2$ and divide the length into three equal strips, each with $A_z = A/3 = 2.1677\,\mathrm{m}^2$. The bench and structure occupy part of this footprint; A is a geometric reference, not a net standing area or a capacity estimate.

Representative coordinates are $(x_z, W/2)$, with $x_z = (z - \frac{1}{2})L/3$. Delivery is assumed to originate at $x = 0$. Geometry defines the accounting regions; it does not determine the assumed speeds and mixing fractions.

2.1 Data, interpretation, and a reproducible scenario

The problem gives apparent cooling up to 3 °C for A and air-temperature reductions up to 5/8 °C for B/C [5]. We do not equate these quantities or treat A's unspecified "feels like" effect as measured SET.

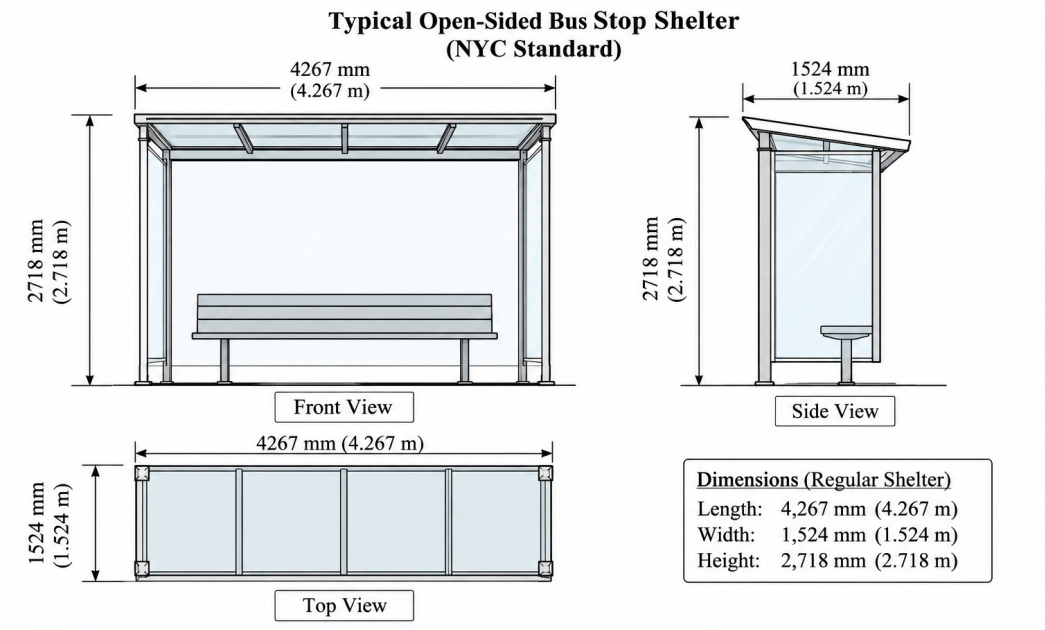


Figure 3: Adopted baseline geometry: regular-size New York City bus shelter. User-supplied schematic; dimensions verified against NYC DOT's *Street Design Manual* [11]. Millimetre labels are rounded.

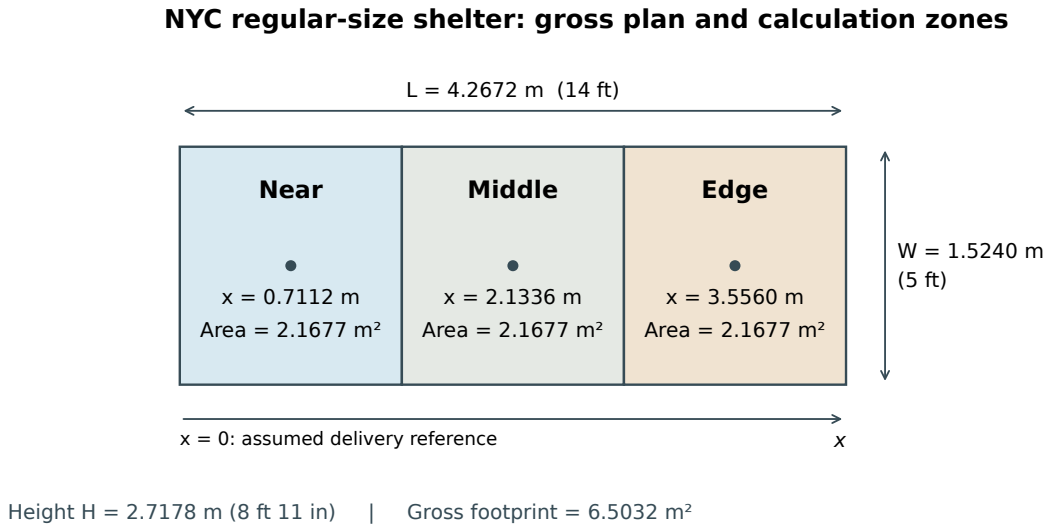


Figure 4: Calculation zones derived from the adopted dimensions. Equal strips and the delivery reference at $x = 0$ are modeling choices; dots mark representative zone coordinates, not passenger locations. No bench or obstruction dimensions are inferred.

We interpret the table's kW/hr entries as mean powers $p_A = 0.25$, $p_B = 1.2$, $p_C = 2.5$ kW; a one-hour comparison uses $0.25/1.2/2.5$ kWh. This is a unit interpretation, not verified equipment specifications.

Airflow, coverage, rating weather and evaporative effectiveness are unspecified. Tables 2–3 define a reproducible scenario rather than identify actual device performance.

Table 2: Common inputs: documented geometry and explicit scenario assumptions.

Quantity	Value, basis, and interpretation
Shelter	NYC regular-size geometry: $4.2672 \times 1.5240 \times 2.7178$ m [11]; three equal $2.1677\,\mathrm{m}^2$ gross zones. Full shade is assumed.
Weather	$T_a = 34\,^\circ$C; dry/humid cases RH = 30/70%; sweep 20–80%.
Radiation and background air	Mean radiant temperature $T_r = 36\,^\circ$C; baseline local speed $v_0 = 0.20$ m/s; pressure 101325 Pa.
Reference passenger	Standing, 1.2 met, 0.5 clo; default body area $1.8258\,\mathrm{m}^2$ and zero external work in the SET implementation.
Occupancy and period	Assumed mean simultaneous occupancy $(4, 4, 4)$ over one hour: 12 person-hours; an illustrative load, not a measured count or rated capacity.
Alternative distributions	Concentration stress cases $(8, 3, 1)$ and $(1, 3, 8)$; both retain the same total of 12 people.
Decision threshold	SET threshold $\Theta_c = 26\,^\circ$C, an assumed hot-discomfort reference; tested at 24 and 28 °C.

Table 3: Assumed local delivery profiles. Entries follow near/middle/edge order.

Device	Local air speed (m/s)	Treated-stream fraction $\beta_{s,z}$	ε_s
A	1.00/0.55/0.25	immaterial at $\chi = 0$	—
B	0.80/0.50/0.30	0.80/0.45/0.10	0.70
C	1.00/0.80/0.50	0.90/0.70/0.40	0.70

The entries represent zone averages over the strips in Figure 4; they are not measured NYC delivery data. Here $\beta_{s,z}$ is a local dry-air mixing fraction, not the share of total device flow assigned to a zone; its entries need not sum to one. The larger coverage assumed for C is tested, not presented as a fact supplied by the problem.

2.2 Predicting the local environment

Fan A: air movement. We set $T_{A,z} = T_a$ and retain ambient moisture, while prescribing local speeds from Table 3. The resulting convection and evaporation enter the passenger-response model. This follows the physical distinction between moving air and lowering dry-bulb temperature [7]. The introductory bus calculation separately neglects local motor/pump heating. A fixed outdoor background alone does not justify that approximation; we relax it in the heat-allocation sensitivity and in the rail extension.

B and C: evaporation followed by dilution. Direct evaporation reduces dry-bulb temperature while increasing moisture; the entering wet-bulb temperature T_{wb} limits the achievable depression [6]. We adopt

$$T_{o,s} = T_b - \min\{\Delta T_s^{\max},\ \varepsilon_s(T_b - T_{wb,b})\}, \qquad s \in \{B, C\}, \tag{5}$$

where the inlet/background state b is outdoor air for the bus and evolving local background air for the rail station. Here $\Delta T_B^{\max} = 5$ and $\Delta T_C^{\max} = 8\,^\circ$C. Applying these advertised shelter-temperature reductions

as outlet caps is an explicit modeling convention; the problem does not specify their measurement location. The caps do not imply that either device achieves them under every climate.

Let w denote kg of water vapor per kg of dry air and h the moist-air enthalpy in J/kg of dry air. Using standard SI psychrometric relations [10],

$$w = 0.621945\frac{p_v}{p - p_v}, \qquad p_v = \phi p_{sat}(T), \tag{6}$$

$$h(T, w) = 1006T + w(2501000 + 1860T), \tag{7}$$

where ϕ is RH as a fraction. Approximating evaporation as isenthalpic, $h_{o,s} \simeq h_b$, gives

$$w_{o,s} = \frac{h_b - 1006T_{o,s}}{2501000 + 1860T_{o,s}}. \tag{8}$$

This neglects fan heat and make-up-water enthalpy; near-constant enthalpy is not an assertion of exact constant wet-bulb temperature [4].

One local-delivery model for both venues. Let a device stream $\dot{m}_{s,z}$ mix with entrained background air $\dot{m}_{e,z}$ in a small region around the passenger. With negligible local air storage, balanced outflow, and $x \in \{h, w\}$,

$$0 = \dot{m}_{s,z}(x_s^{\text{sup}} - x_{\ell,s,z}) + \dot{m}_{e,z}(x_b - x_{\ell,s,z}), \qquad \beta_{s,z} = \frac{\dot{m}_{s,z}}{\dot{m}_{s,z} + \dot{m}_{e,z}}. \tag{9}$$

A fraction χ of device electricity may dissipate into the supply air before reaching people. Write p_s for unit power in kW and $P_s = 1000p_s$ for watts. With unit dry-air supply flow $\dot{m}_s$, define

$$h_s^{\text{sup}} = h_{o,s} + \chi P_s/\dot{m}_s, \qquad w_s^{\text{sup}} = w_{o,s}, \qquad 0 \le \chi \le 1. \tag{10}$$

For a fan, the unheated outlet equals its inlet. Solving Eq. (9) gives the shared operator

$$w_{\ell,s,z} = (1 - \beta_{s,z})w_b + \beta_{s,z}w_s^{\text{sup}}, \tag{11}$$

$$h_{\ell,s,z} = (1 - \beta_{s,z})h_b + \beta_{s,z}h_s^{\text{sup}}, \tag{12}$$

$$T_{\ell,s,z} = \frac{h_{\ell,s,z} - 2501000w_{\ell,s,z}}{1006 + 1860w_{\ell,s,z}}. \tag{13}$$

RH is recovered from T_ℓ, w_ℓ, not averaged. All local states must satisfy psychrometric and SET-domain limits. The bus base case uses $b = a$ and $\chi = 0$. Its assumed β values imply entrainment-to-supply ratios $(1 - \beta)/\beta$: 0.25/1.22/9.00 for B and 0.11/0.43/1.50 for C. These are an explicit flow-ratio closure, not newly measured airflow. If the strips are supplied in parallel, their supply branches must sum to one device's flow; no full-device flow is counted independently in every strip.

The local quasi-steady approximation requires air residence time to be short relative to weather or operating changes. No measured residence time is available, so the bus calculation is a conditional steady-state comparison. Its fixed outdoor reservoir also omits accumulated crowd heat and humidity. The station extension releases that approximation while retaining Eqs. (5)–(13) and the response below.

2.3 Converting local conditions to a common human response

We compute standard effective temperature (SET), denoted Θ, from

$$\Theta_{s,z} = F_{\text{SET}}(T_{\ell,s,z}, T_{r,z}, v_{s,z}, \phi_{\ell,s,z}; \text{met}, \text{clo}). \tag{14}$$

SET compares physiological heat exchange with a reference environment, incorporating temperature, humidity, air movement and radiation [8, 7]. We use `pythermalcomfort` with applicability checks [9]. SET is a quasi-steady engineering proxy here; it does not resolve each passenger's prior walking or transient physiology.

2.4 Measuring service and retaining a quality check

For the warm-weather scenarios, define the hot-discomfort proxy

$$D(\Theta) = [\Theta - \Theta_c]_+, \qquad [x]_+ = \max(x, 0). \tag{15}$$

The threshold is a decision assumption, not an ASHRAE safety limit. Let 0 denote the same shelter without active cooling, retaining the same shade, weather, radiation, and background air movement. The signed local improvement is

$$b_{s,z,k} = D(\Theta_{0,z,k}) - D(\Theta_{s,z,k}). \tag{16}$$

Negative values are retained: an adverse condition must reduce useful service. Over time steps of Δt_k hours,

$$U_s = \sum_k \sum_z n_{z,k} b_{s,z,k} \Delta t_k, \qquad E_s = \sum_k p_{s,k} \Delta t_k, \tag{17}$$

$$\eta_s = U_s / E_s, \qquad E_s > 0. \tag{18}$$

The unit of η is person·°C·h/kWh, with temperature measured on the SET scale. This is a service-efficiency index, not a thermodynamic percentage or COP. Empty zones earn no credit, although operating electricity is still counted. The zero-energy baseline has no efficiency ratio.

We also report remaining discomfort,

$$R_s = \sum_k \sum_z n_{z,k} D(\Theta_{s,z,k}) \Delta t_k, \qquad \overline{D}_s = \frac{R_s}{\sum_k \sum_z n_{z,k} \Delta t_k}. \tag{19}$$

A high η_s alone does not establish adequate relief. If a decision-maker specifies an acceptable residual or an individual exposure requirement, it should be imposed separately; no universal heat-health threshold is inferred from our SET proxy. When the baseline already meets the chosen hot-discomfort criterion, further cooling earns no benefit, and switching off can be appropriate.

Geometry and demand. Use either headcounts n_z or gross-area densities ρ_z with $n_z = \rho_z A_z$. Area is counted once, not multiplied into the numerator again. The uniform and eight-person-strip cases imply 1.85 and 3.69 persons per gross square metre; these are concentration stress scenarios, not demonstrated usable capacities. Bench and circulation dimensions are needed for a deployable layout. Fixed headcounts and fields give fixed service; fixed densities and fields give service proportional to area. These are accounting properties, not a claim that airflow is independent of geometry.

A ranking result independent of the assumed delivery maps. Under the interpreted constant powers and a common operating period,

$$\begin{aligned} \eta_B > \eta_A &\iff U_B > 4.8 U_A, \\ \eta_C > \eta_A &\iff U_C > 10 U_A, \\ \eta_C > \eta_B &\iff U_C > \frac{25}{12} U_B. \end{aligned} \tag{20}$$

Thus nominal temperature reduction alone cannot justify a larger device. Its accumulated passenger relief must compensate for its larger electricity input. These inequalities follow from the given power ratios; their validity does not depend on the assumed zone profiles.

2.5 Numerical results: local mechanisms and the efficiency tradeoff

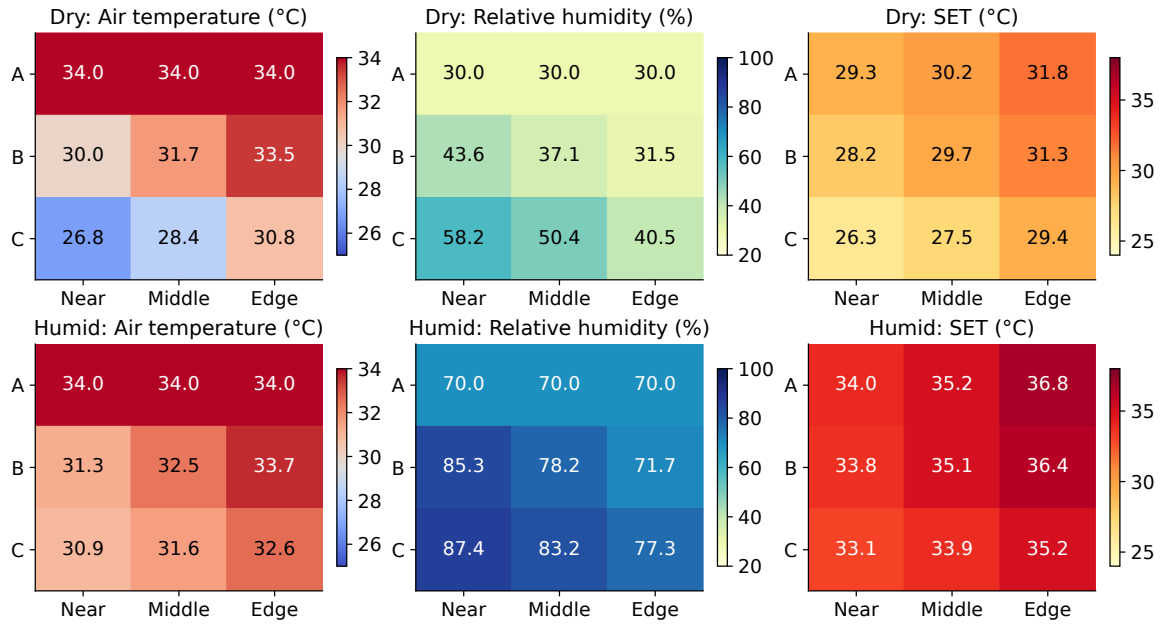


Figure 5: Simulated local air temperature, RH, and SET under the stated assumptions. Fan A leaves air temperature unchanged while changing SET. B and C lower air temperature and increase RH. These are scenario outputs, not field measurements.

Spatial delivery matters even within one small shelter. In the dry case, C gives SET 26.3 °C near the outlet but 29.4 °C at the edge. At RH 70%, these become 33.1 and 35.2 °C. A single mean air-temperature reduction would conceal these differences. A, meanwhile, improves SET without changing the 34 °C dry-bulb temperature.

For uniform occupancy, dry-case efficiencies are 84.6, 24.5, and 21.1 for A, B, and C, respectively. Yet their mean remaining SET excesses are 4.40, 3.72, and 1.76 °C: C delivers more total relief while A delivers more per kWh. At RH 70%, efficiencies are 90.6, 21.1, and 15.2; even C retains a mean excess of 8.05 °C. The efficiency winner therefore does not establish that the shelter meets a comfort or safety requirement.

2.6 Climate and passenger distribution

B/C's assumed outlet drops are 5/8 °C at RH 30% but approximately 3.4/3.4 °C at RH 70%. Equal high-humidity drops follow from the common assumed effectiveness once neither cap binds; delivery profiles still differ. Nonlinear SET and a changing baseline mean service efficiency need not vary monotonically with RH.

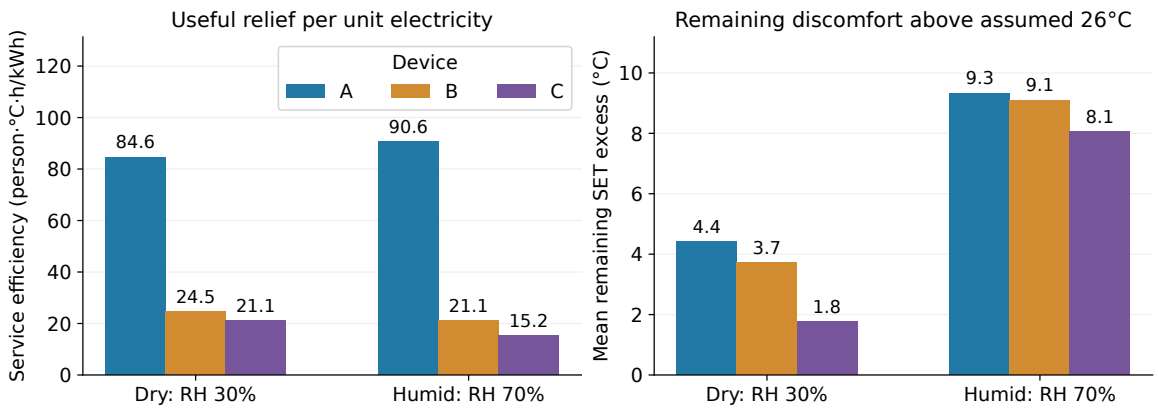


Figure 6: Uniform occupancy: electricity-normalized relief and remaining hot-discomfort proxy. Lower residual is better, whereas higher service efficiency is better. The 26 °C SET reference is assumed.

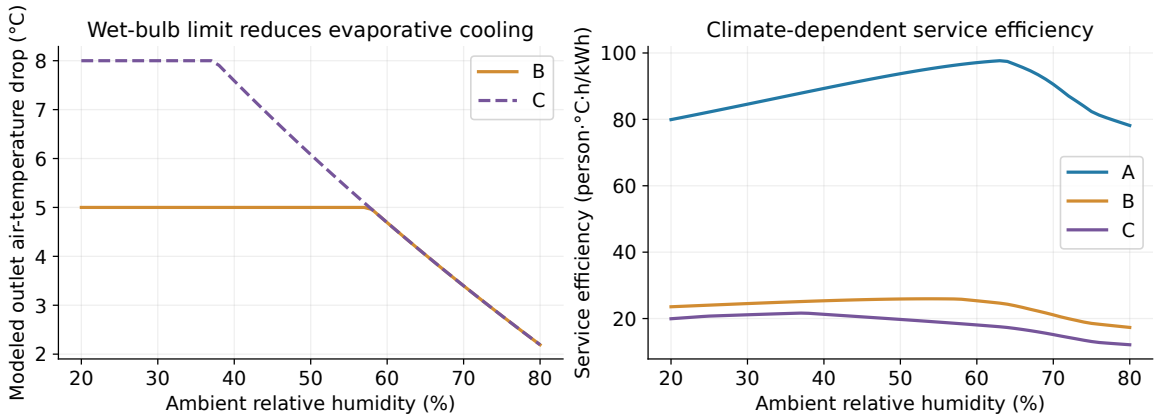


Figure 7: RH sweep with temperature, radiation, occupancy, and delivery profiles fixed. Outlet cooling is bounded by both the specified cap and the wet-bulb depression.

In dry conditions, A's efficiency falls from 118.3 with near-heavy occupancy to 48.3 with edge-heavy occupancy. B exceeds C under uniform occupancy (24.5 versus 21.1), but C exceeds B under edge-heavy occupancy (16.6 versus 15.5). A remains the efficiency leader in all six nominal climate–occupancy cases. Full occupancy and endpoint-sensitivity figures remain separately available with the computational material.

2.7 When the recommendation changes

A poorly aimed or obstructed fan can consume electricity without delivering its nominal air movement to passengers. We test this specific failure mode using

$$v_{A,z}(q) = v_0 + q(v_{A,z}^{\mathrm{nom}} - v_0), \qquad 0 \le q \le 1, \tag{21}$$

while holding $p_A = 0.25$ kW. Here q describes delivery loss at occupied positions, *not* a fan-speed control setting; changing motor speed would require a different power model. The B/C profiles stay fixed in this particular stress test.

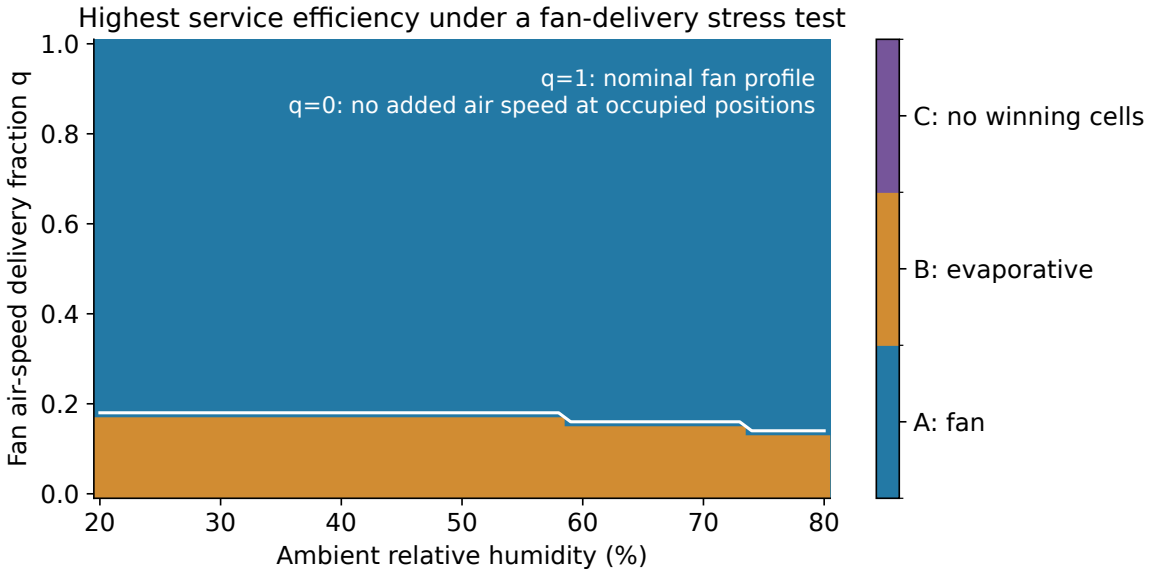


Figure 8: Conditional efficiency winner on the complete RH–fan-delivery grid: RH 20–80% in one-percentage-point steps and $q = 0$–1 in steps of 0.02. The boundary is grid-resolved. This tests fan delivery loss; it is not a universal climate-selection map.

The first sampled q at which A wins is 0.18 at RH 30% and 0.16 at RH 70%; the switch lies between that sample and the preceding 0.02 step. B wins below these levels; C wins no cell of this specific grid. These are conditional delivery thresholds, not universal climate recommendations. Measure occupied-location (T, RH, v, T_r), electricity and occupancy to replace the assumed profiles before using the comparison for an installation.

2.8 Sensitivity, verification, and scope

Across the tested individual endpoint changes, the largest B/A and C/A efficiency ratios are 0.432 and 0.372; no endpoint reverses the nominal advantage of A. This does not prove robustness over every intermediate value or joint parameter combination. Added-speed factors scale $v - v_0$, and cooled-air fractions are capped at one. The rating-interpretation test removes the 5/8 °C outlet caps while retaining the wet-bulb constraint; it addresses uncertainty in where the quoted temperature effects were measured. The delivery-loss test in Figure 8 already demonstrates that stronger departures can change the winner.

Relaxing the bus heat approximation. A separate $\chi = 0, 0.5, 1$ test assumes unit supply flows 0.6/0.6/1.2 kg dry air/s and, for A's previously immaterial scalar mixing, $\beta_A = (0.8, 0.45, 0.1)$. These additional assumptions are not inferred from rated power. From $\chi = 0$ to 1, dry-case A/B/C efficiencies change from 84.6/24.5/21.1 to 81.0/20.8/18.0; humid-case values change from 90.6/21.1/15.2 to 86.8/17.4/12.1. A remains the most efficient and C leaves the lowest residual in these cases, but

the humid-case A/B residual ordering changes. This identifies a consequence of the introductory zero-local-heat approximation; the rail comparison explicitly includes this heat and tests its allocation.

Verification. The implementation checks psychrometric round trips, dry-air moisture/enthalpy mixing balances, wet-bulb bounds, and saturation. Service-accounting checks verify $U_s + R_s = R_0$, zero benefit at zero occupancy, zero benefit when all delivered effects vanish, and unchanged rankings when every zone's population doubles. A signed-adverse-effect check confirms that worsening discomfort is not clipped away. No credit is given for additional cooling of an already comfortable baseline. Geometry checks verify unit conversion and area conservation, equivalence of count and density inputs, fixed-count invariance, and fixed-density scaling under unchanged local fields.

Limitations and transfer. Local profiles and representative passengers remain assumptions; the numerical ranking is conditional on them. The conservation-based mixing closure uses assumed flow ratios; fan curves, crosswind and opening geometry have not been used to identify them. Radiation is fixed for a shaded shelter, so direct sun and changing shadow patterns need additional modeling. Short waiting times and varied physiological histories are not resolved by quasi-steady SET. Without measured flow, water consumption cannot be calculated reliably from the moisture increase alone; water and operating cost belong in the later effectiveness analysis. Section 3 retains the same device, local mixing, SET and service modules, and adds a dynamic background, surface storage and variable equipment deployment.

Reproducibility. The project retains PsychroLib 2.5.0 and `pythermalcomfort` 4.8.1 inputs, source code, CSV results, software versions and verification records. Each figure is a separately replaceable PDF. These are conditional simulations, not experimental validation.

3 Canberra MRT Station: A Semi-open Waiting-area Case

3.1 Site evidence and the modeled waiting area

We select Canberra MRT station in Singapore for Requirement 2(b). RSP describes an elevated, two-storey station with a site area of 9, 128 m^2 [16]. LTA identifies platform planting, daylight-admitting glazing and rain-protective louvers that allow natural ventilation [17]. The design question is which device types, quantities and locations can meet passenger cooling and coverage requirements with the least electricity. We evaluate candidate installations of A, B and C in this shaded, humid, semi-open setting; their deployment is hypothetical.

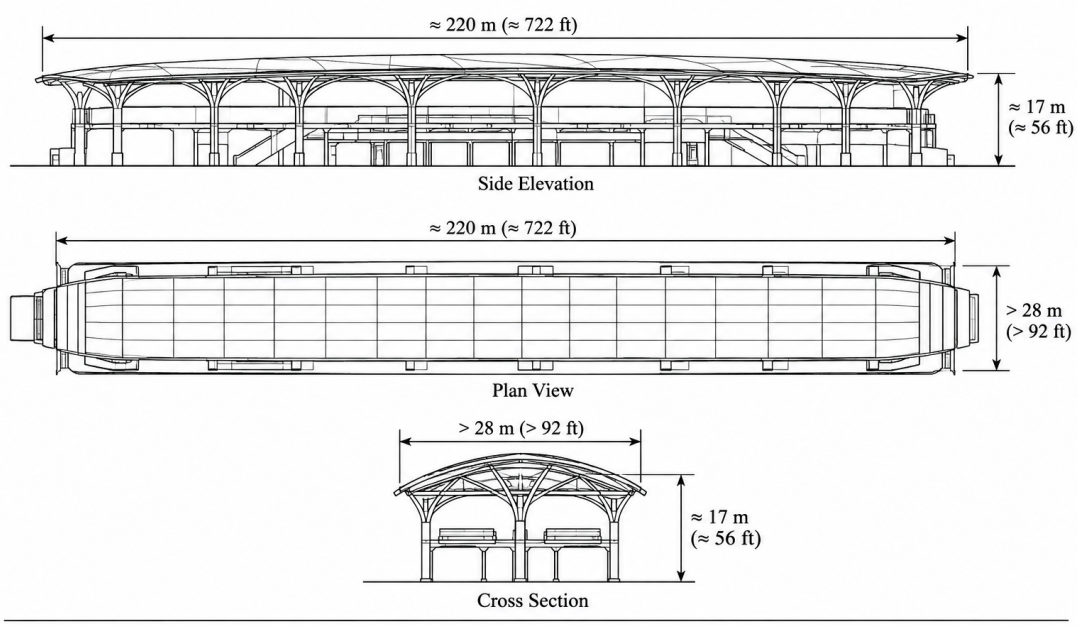


Figure 9: User-supplied Canberra station schematic, retained as a separate PDF. Approximate structural dimensions are contextual references; the drawing is not an as-built survey or a map of the calculation zones. Sources identified in the drawing are listed in [18, 16].

Modeled occupied area. The schematic labels approximately 220 m length, roof span over 28 m and roughly 17 m columns. Roof span and column height are not occupied width or platform clearance. We assume aggregate waiting width $b = 6$ m, usable fraction $\alpha = 0.8$ and effective background-air height 4 m:

$$A_{\text{occ}} = \alpha L b = 1056\,\text{m}^2. \tag{22}$$

Three equal aggregate zones, access/core/edge, each have $352\,\text{m}^2$ and $1408\,\text{m}^3$. They represent contrasting exposure conditions, not surveyed compartments. Neither occupied area nor effective height is inferred directly from site area or column dimensions.

3.2 What changes from the bus model, and why

The service objective, device response, local mixing operator, SET and energy denominator remain unchanged. What changes is the background state supplying each device. The bus uses a prescribed outdoor reservoir; the station solves Eq. (4) with time-varying people, exchange and surface feedback. Local delivered air is then computed using the same Eq. (13). This makes the extension traceable at both the physical and service levels.

Table 4: Retained modules, released approximations, and their consequences.

Module	Bus simplification	Rail extension and purpose
Device and local delivery	Fixed outdoor inlet; steady entrainment; local electric heat neglected.	Evolving background inlet; same mixing balance with supply heat, to distinguish local relief from background response.
Background heat/moisture	Outdoor reservoir is prescribed.	Solve storage and interzone exchange to capture humidity feedback and time lag.
Surfaces	Fixed shaded radiant environment.	Solve effective surface storage and radiative influence.
Response and service	SET, signed relief, people-time weighting and all-device energy.	Retain all four; vary counts and locations and assess coverage.

The assumed base exchange of 4–10 h^{-1} gives an air replacement time $(3600/ACH)$ s = 6–15 min, comparable to the assumed 15-min passenger cycle. This supports resolving background dynamics in this scenario; station size alone would not justify it. The local mixing region is treated as having negligible storage only if $\tau_\ell = M_\ell/(\dot{m}_k + \dot{m}_e)$ is small relative to the 120-s arrival disturbance. M_ℓ is the air held in the local jet core, not footprint area multiplied by the platform height. Its unmeasured residence time is a limitation of this closure.

3.3 Coupled heat, moisture, and surface balances

Following the control-volume structure of building heat and moisture balances [13], let each zone store moist-air enthalpy h_z (J/kg dry air), humidity ratio w_z (kg/kg dry air), and an effective roof/wall inner-surface temperature $T_{m,z}$. The background air is well mixed within each aggregate zone; the embedded local delivery regions are evaluated separately with negligible air storage. Background dry-air mass is approximated as constant, $M_z = \rho_{da,o} V_z$, with outdoor reference density. Using the occupied-zone volume for background storage neglects the smaller local jet hold-up. This is a reduced model, not an EnergyPlus or CFD simulation.

Define the exchange operator, for either $x = h$ or w, as

$$\mathcal{L}_z(x) = \dot{m}_{0z}(x_o - x_z) + \sum_{j \neq z} \dot{m}_{zj}(x_j - x_z), \qquad \dot{m}_{0z} = M_z \lambda_z / 3600. \tag{23}$$

Here λ_z is the assumed outdoor air-change rate in h^{-1}. Each exchange represents equal entering and leaving dry-air flow; adjacent-zone rates satisfy $\dot{m}_{zj} = \dot{m}_{jz}$. Wind, buoyancy and opening geometry can affect outdoor exchange [15]; a prescribed train-cycle pulse tests that uncertainty without claiming a station-specific piston-flow law.

With x_{zk} devices of type k in zone z, unit circulation $\dot{m}_k$ (kg dry air/s), and time in seconds, our balances are

$$M_z \dot{h}_z = \mathcal{L}_z(h) + H_{ma,z}(T_{m,z} - T_z) + n_z q_c + n_z g h_v(T_z) + P_z + \sum_k x_{zk} \dot{m}_k (h_{e,zk} - h_z), \tag{24}$$

$$M_z \dot{w}_z = \mathcal{L}_z(w) + n_z g + \sum_k x_{zk} \dot{m}_k (w_{e,zk} - w_z), \tag{25}$$

$$C_{m,z} \dot{T}_{m,z} = H_{mo,z}(T_o - T_{m,z}) + Q_{\mathrm{sol},z} + n_z q_r - H_{ma,z}(T_{m,z} - T_z). \tag{26}$$

The per-person inputs are convective heat q_c, radiative heat q_r, and water-vapor generation g. Vapor enters with $h_v(T) = 2501000 + 1860T$ J/kg; no additional latent-heat term is added. With $P_k = 1000 p_k$ W per unit, total device power is $P_z = \sum_k x_{zk} P_k$ (W). This power ultimately dissipates within the modeled zone. Its division between supply air and background affects local exposure but not total input. Pump power is already included in the device total. Conductances H have units W/K, surface capacity C_m has units J/K, and Q_{sol} is an effective absorbed envelope load. The opposite signs of the air–surface term conserve internal energy.

Air temperature is recovered from $T_z = (h_z - 2501000 w_z)/(1006 + 1860 w_z)$. A shaded longwave approximation for the surface contribution to comfort [8] gives

$$T_{r,z} = \left[F_z (T_{m,z} + 273.15)^4 + (1 - F_z)(T_{r,o} + 273.15)^4 \right]^{1/4} - 273.15. \tag{27}$$

F_z is an assumed effective view fraction; the example sets external radiant temperature $T_{r,o} = T_o$, while keeping it a separate input. Direct sun on passengers and detailed radiative asymmetry are excluded.

The retained device module. Evaluate Eq. (5) using the current background inlet. For A, the unheated circulation outlet equals the background, so its circulation has no net bulk enthalpy or water source. For B/C, $h_{e,zk} \simeq h_z$ while $w_{e,zk} > w_z$ when evaporation occurs. Increasing background moisture reduces subsequent wet-bulb depression. Here the subscript e denotes the unheated outlet, the state labeled o, s in Eqs. (5)–(7); it does not denote outdoor air. Device electricity remains a separate positive heat source in Eq. (24).

3.4 Local delivery without double-counting heat or water

For each covered subgroup, apply Eqs. (10)–(13) with background $b = z$. We distinguish the treated-stream fraction β_{zk} from the passenger coverage fraction c_{zk} defined below. The supply flow draws from the background, entrains additional background air, and returns the mixed air to that same zone. For finite entrainment ($\beta > 0$),

$$(\dot{m}_k + \dot{m}_e)(h_{\ell,zk} - h_z) + (1 - \chi) P_k = \dot{m}_k (h_{e,zk} - h_z) + P_k, \tag{28}$$

$$(\dot{m}_k + \dot{m}_e)(w_{\ell,zk} - w_z) = \dot{m}_k (w_{e,zk} - w_z). \tag{29}$$

Thus the return flow gives exactly the existing background source. The local state improves the exposure calculation without adding evaporation or electrical heat a second time. People outside direct footprints experience the background state; those inside experience the mixed supply state. Both experience the zone radiant environment. Prescribed air speeds remain a separate approximation; no momentum or detailed jet trajectory is solved.

We set $\beta = 0.5$ for each type as an explicit equal-entrainment scenario and use unit circulation flows 0.6/0.6/1.2 kg dry air/s for A/B/C. These are assumptions, not manufacturer data. The nominal $\chi = 1$ puts all electrical dissipation into the air before it reaches passengers, a conservative supply-heating case. The sensitivity analysis varies β and χ, and separately varies A's newly assumed circulation flow. The case $\beta = 0$ is an infinite-dilution scalar limit that exactly recovers the previous shared-background exposure; it retains the prescribed local speeds and is a diagnostic limit, not a measured finite-flow state.

Recovery of the bus model. Hold the background at outdoor h_a, w_a, prescribe the bus radiant environment, speeds and occupancies, neglect local electrical heating ($\chi = 0$), and use the bus β values. The same device and delivery functions then give the bus temperatures, humidities, SET and service exactly. Fixing the background corresponds to an outdoor-reservoir approximation to Eq. (4); merely setting time derivatives to zero would not impose it. This distinction makes the reduction physically explicit.

3.5 Equipment selection, allocation, and coverage

Here s indexes an installation and k a device type. Let $x_{zk} \in \mathbb{Z}_{\geq 0}$ be the number of devices of type $k \in \{A, B, C\}$ in zone z. A 3×3 count matrix specifies a candidate installation, allowing quantities and device types to vary by zone. Each candidate is evaluated through the resulting shared thermal environment: all operating units contribute heat and electricity, and evaporative units also contribute moisture.

We use assumed, non-overlapping service footprints $a_A = 20$, $a_B = 30$ and $a_C = 40\,\mathrm{m}^2$ per unit. These are comparison inputs, not manufacturer coverage ratings. For uniform passenger density within a zone, define

$$c_{zk} = \frac{x_{zk} a_k}{A_z}, \qquad c_{z0} = 1 - \sum_k c_{zk} \geq 0, \qquad \sum_k x_{zk} a_k \leq A_z. \tag{30}$$

The uncovered fraction experiences local speed v_0; each covered subgroup experiences its device-specific speed v_{zk}. We average *discomfort*, not velocities or device efficiencies:

$$\overline{D}_{s,z}(t) = c_{z0} D[\Theta(T_z, T_{r,z}, v_0, \phi_z)] + \sum_k c_{zk} D[\Theta(T_{\ell,zk}, T_{r,z}, v_{zk}, \phi_{\ell,zk})]. \tag{31}$$

The subgroups share radiant conditions but can differ in air temperature, humidity and speed. An uncovered passenger can still experience indirect background cooling or humidification. Coverage denotes direct delivery, not a boundary of all thermal influence. Separate, non-overlapping footprints approximate distributed equipment; co-located jets, overlap and series operation require additional airflow modeling.

3.6 Canberra scenario and comparable deployment tests

We assume 33 °C and 70% outdoor RH for a hot, humid Canberra scenario. Singapore's weather service describes typical daytime maxima of 31–33 °C and high humidity, with rain-free afternoon RH around 60% [19]. Our pair is a scenario, not a simultaneous observation at the station. SET retains 1.2 met, 0.5 clo, and the common 26 °C reference.

Table 5: Station-scale scenario: operating assumptions, with powers and outlet caps interpreted from the problem.

Input	Assumed value and interpretation
People and train cycle	An assumed 15-min stress-test cycle, not the actual timetable. Zone counts rise from $(26.4, 44, 17.6)$ to $(88, 198, 66)$, then reset; fractional values represent mean counts.
Outdoor and zone exchange	Base ACH $(8, 4, 10)$ h^{-1}, plus $(8, 4, 12)$ in the last 2 min; adjacent-zone exchange 0.88 kg dry air/s.
Surface response, per zone	$C_m = 17.6$ MJ/K; $H_{ma} = 880$, $H_{mo} = 1760$ W/K; absorbed load 3520 W; view fractions $(0.55, 0.85, 0.55)$.
Human sources	Convection 45 W, radiation 30 W, vapor $45/(2.45 \times 10^6)$ kg/s per person.
Device powers and flows	$p_{A/B/C} = 0.25/1.2/2.5$ kW; circulation $0.60/0.60/1.20$ kg dry air/s per unit. B/C use $\varepsilon = 0.70$ and $5/8$ °C outlet-drop caps. Local $\beta = 0.5$, $\chi = 1$.
Speeds at covered locations	A $(0.60, 1.00, 0.50)$, B $(0.50, 0.80, 0.45)$, C $(0.75, 1.00, 0.60)$ m/s; uncovered locations 0.20 m/s.

The same occupied area, thermal properties, load densities and passenger schedule are used for every candidate. This holds demand constant while equipment quantities and allocation change. The architectural sources establish the venue context; the assumed operating conditions and coverage parameters define the numerical comparison.

We first compare six benchmark configurations at the same direct-service footprint: 240 m^2 per zone, or 68.2% of the assumed occupied area. Per-zone counts (x_A, x_B, x_C) are $(12, 0, 0)$, $(0, 8, 0)$, $(0, 0, 6)$, $(6, 4, 0)$, $(6, 0, 3)$ and $(3, 2, 3)$. Whole-station totals are respectively 36, 24, 18, 30, 27 and 24 units. Device-count sweeps then quantify the energy needed for wider coverage, and a fixed-inventory redistribution test isolates the effect of allocation. Together these comparisons address type, quantity and placement.

A prescribed two-hour uncooled warm-up gives all candidates the same initial air and surface states. Each then runs for two hours alongside a no-device trajectory with identical weather, people and exchange. This specifies common prior history without claiming periodic steady state; no free initial stored cooling is credited. Electricity covers all cooling devices, not the whole station bill; benefits after the horizon are excluded.

3.7 Comparing service, coverage, and electricity

Retain signed relief against the matched baseline, now using the coverage-weighted discomfort:

$$U_s = \frac{1}{3600} \int_0^{7200} \sum_z n_z(t) [D_{0,z}(t) - \overline{D}_{s,z}(t)] \, dt, \tag{32}$$

$$E_s = \frac{1}{3600} \int_0^{7200} \sum_{z,k} x_{zk} p_k \, dt, \qquad \eta_s = U_s / E_s. \tag{33}$$

With t in seconds and p_k in kW, the factors $1/3600$ convert to hours. The units remain person·°C·h/kWh. Report mean remaining discomfort $\overline{R}_s = \int \sum_z n_z \overline{D}_{s,z} \, dt / \int \sum_z n_z \, dt$ alongside η_s and coverage. Se-

lection proceeds in two steps: retain candidates that meet the specified remaining-discomfort and person-hour coverage requirements, then choose the lowest-energy candidate. Because acceptance thresholds have not been specified for this scenario, we report the energy–discomfort frontier and coverage of the tested candidates. This identifies the available choices while leaving the final device count dependent on service requirements.

The common two-hour baseline contains 440 person-hours and has mean remaining SET excess 10.57 °C. Table 6 reports the revised local-delivery calculation. At equal 68.2% coverage, A36 uses the least electricity and leaves the lowest remaining discomfort among the six nominal benchmarks. Across all 24 candidates, the sampled energy–discomfort frontier consists of A3, A9, A18, A27, A36, A45 and A51. A final count depends on required coverage and residual discomfort; the finite sweep is not a global optimum.

Table 6: Two-hour nominal results ($\beta = 0.5$, $\chi = 1$). Efficiencies use person·°C·h/kWh. All six regular benchmarks have 68.2% direct coverage; redistribution changes person-time coverage.

Plan	Electricity (kWh)	Person-time coverage	Remaining excess (°C)	η
A36	18.0	68.2%	9.12	35.49
B24	57.6	68.2%	9.55	7.75
C18	90.0	68.2%	9.54	5.03
AB30	37.8	68.2%	9.25	15.39
AC27	54.0	68.2%	9.22	10.99
ABC24	63.9	68.2%	9.38	8.21
AB30-R	37.8	76.2%	8.99	18.39

Allocation and coverage. AB30-R relocates the same 18 A and 12 B units from equal zone counts to $(3, 4, 0)$, $(12, 3, 0)$ and $(3, 5, 0)$. Electricity stays at 37.8 kWh and area coverage at 68.2%, while person-time coverage rises to 76.2%. Remaining excess falls from 9.25 to 8.99 °C and efficiency rises from 15.39 to 18.39. This isolates the value of placing a fixed inventory closer to demand. Scaling all footprints by $0.75/1/1.25$ gives A36 remaining excess $9.53/9.12/8.70$ °C. A36 remains the lowest-energy and lowest-residual benchmark at these common footprint multipliers; independently uncertain device footprints are not exhausted by this test.

Why delivery assumptions matter. Across the 15 tested combinations $\beta = 0, 0.25, 0.5, 0.75, 1$ and $\chi = 0, 0.5, 1$, A36 retains the highest efficiency among the six equal-coverage benchmarks. The lowest remaining discomfort is less stable: with $\chi = 0$, AC27 becomes the lowest-residual benchmark for $\beta \geq 0.5$; it also does so at $(\beta, \chi) = (1, 0.5)$. At $(0.5, 0)$ its remaining excess is 9.02 °C, compared with A36's 9.06 °C, but it uses 54 rather than 18 kWh. Thus local delivery can change the quality ordering without changing the efficiency ordering. The result determines a tradeoff for the specified requirements, not a universal preferred equipment type.

Changing A's assumed circulation from 0.3 to 0.6 to 1.2 kg dry air/s gives A36 remaining excess $9.18/9.12/9.09$ °C and efficiency $34.06/35.49/36.22$. A36 keeps the lowest residual and highest efficiency among the six benchmarks at those settings. These checks cover a stated parameter grid; they do not establish robustness to every combination of weather, flow, coverage and passenger distribution.

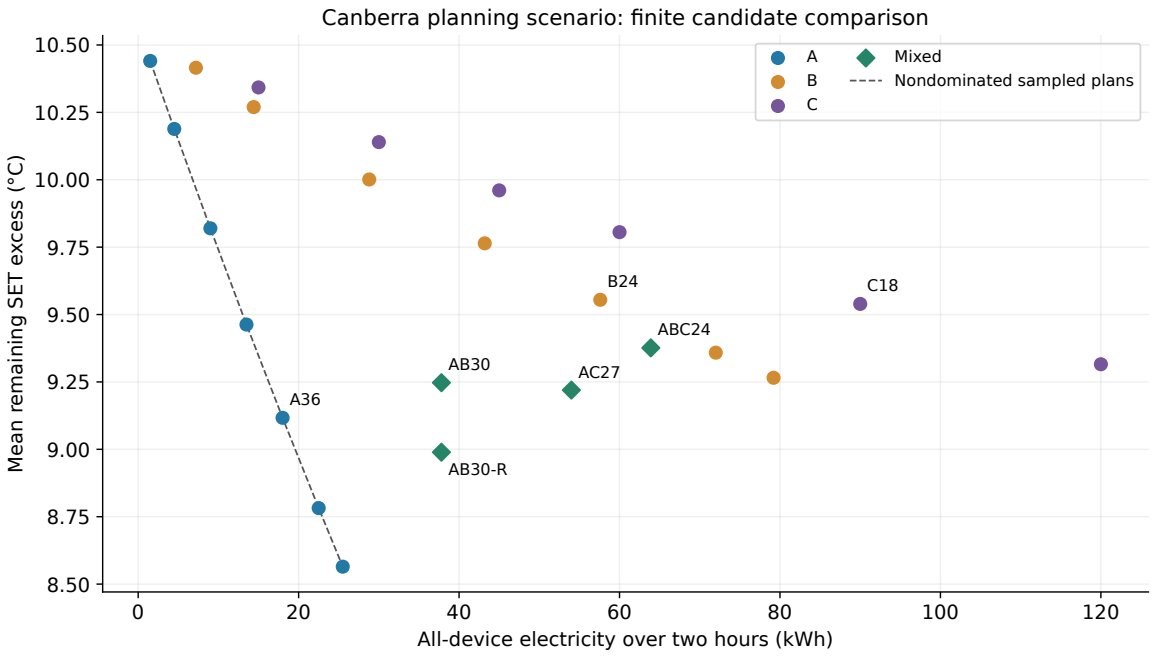


Figure 10: Energy–discomfort tradeoffs for 24 candidate configurations. The dashed line marks the nondominated candidates; required passenger coverage guides selection.

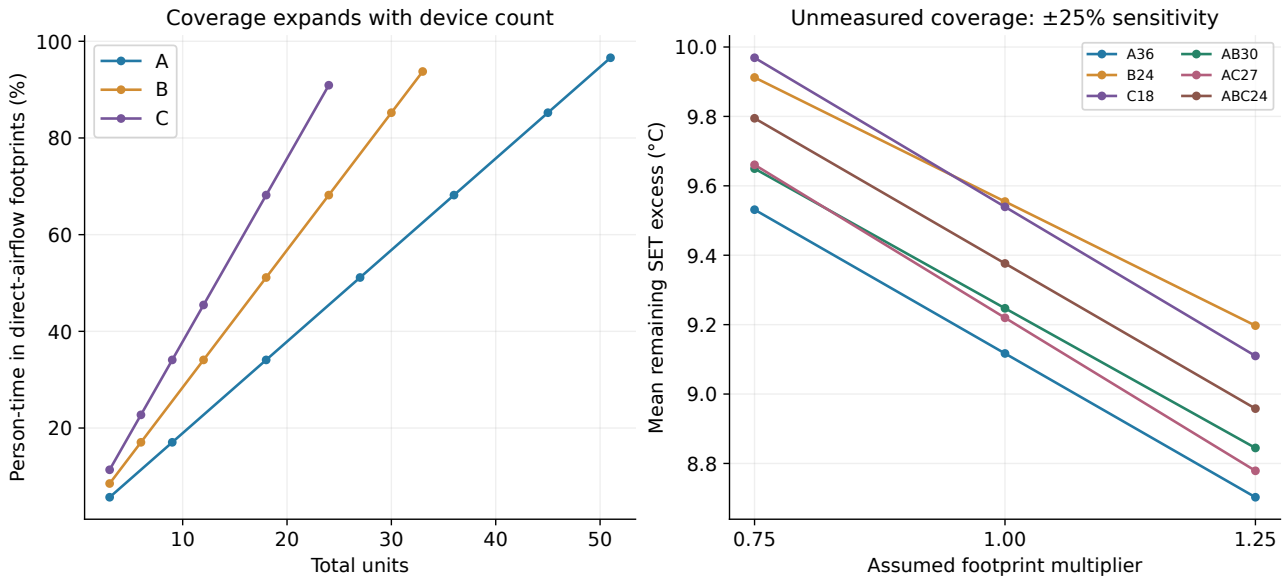


Figure 11: Device-count and assumed-footprint sensitivity. Changing footprint affects direct-service fractions; numerical coverage is not a verified equipment rating.

3.8 Verification, evidence, and transfer

The implementation checks both the full background budgets and the local return identities in Eq. (29). Limiting tests recover the old shared-background station calculation at $\beta = 0$ and the bus calculation

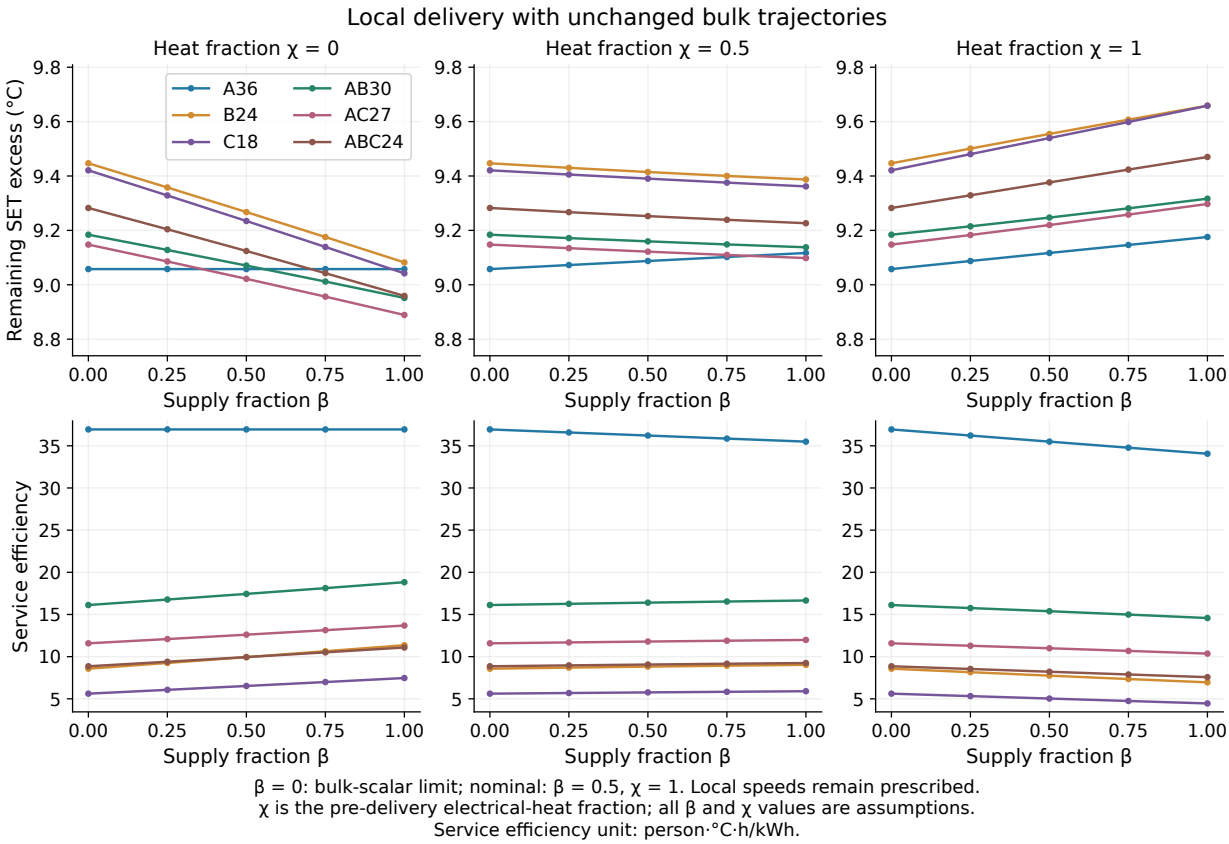


Figure 12: Sensitivity to local stream mixing and supply heat allocation. The parameters are assumptions; $\beta = 0$ recovers shared-background exposure. Rankings are conditional on these delivery choices.

under its fixed-background assumptions. Refining the time step tests integration error; varying β, χ changes local exposure without changing the already balanced background trajectories. The complete check results and trajectories accompany the paper.

The experimental study that informed this structure reports an air-temperature MAE of 0.4 °C and an RH MAE of about five percentage points for its mist-on validation period [14]. It separates calibration and validation periods and compares operation against a matched no-mist simulation. We adopt those comparison principles, but its dry-season, unoccupied cafeteria is not a calibration dataset for Canberra. Its reported average 1.9 °C cooling and mist effectiveness coefficients are not copied into our scenario. It also did not evaluate occupant comfort, so its cooling effectiveness cannot replace our SET-based service measure.

For field validation, first measure the no-device background and estimate exchange and surface response; then measure device outlet and occupied-location T, RH and speed to estimate delivery and heat allocation. Use separate periods to test predictions against matched baseline conditions and metered electricity. Compare occupied positions as well as background averages: the study's own well-mixed-zone limitation motivates our local-delivery refinement. Device footprints, passenger distribution, residence times and flows remain uncertain, so present results are conditional comparisons of sampled installations.

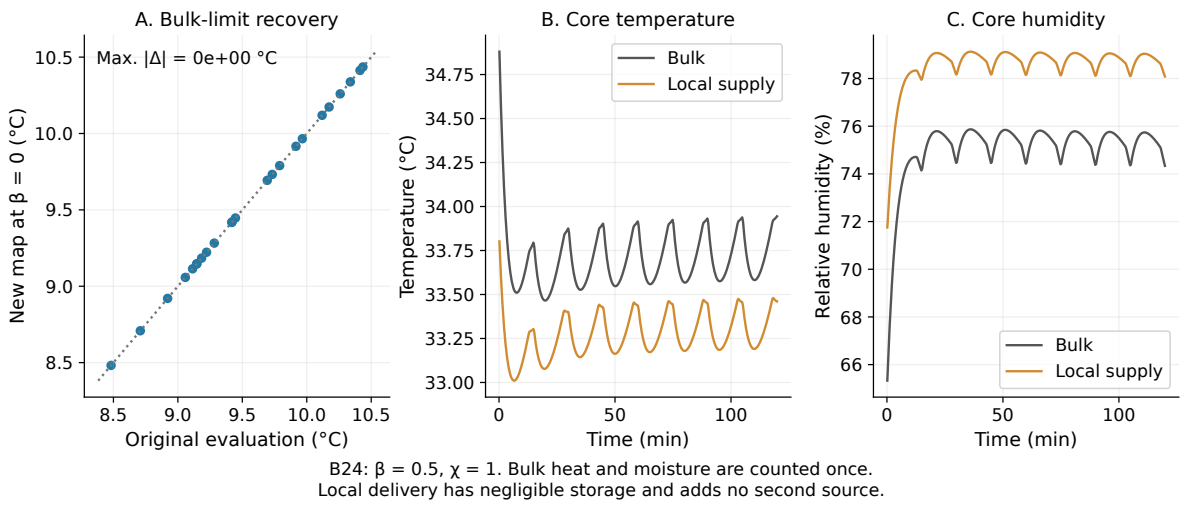


Figure 13: Recovery of the prior station evaluation at $\beta = 0$, and local versus background temperature and humidity for B24. Agreement in the limiting case verifies the implementation, not field accuracy.

The model excludes direct sun, detailed stratification, surface moisture sorption and condensation; it stops on invalid humidity or SET inputs. SET remains quasi-steady even when background conditions evolve. Source code, scenario inputs, numerical checks, CSV outputs and separately replaceable PDF figures are retained in the project. The same chain can be applied to later venues by changing evidenced boundaries and inputs while retaining device, delivery, response and service modules.

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Report on Use of AI Tools

Preliminary record for this partial draft.

OpenAI Codex assisted with translation, model formulation, source checking, Python calculations, English drafting, figures and LaTeX preparation for Requirements 1 and 2 during November 2026. Record the exact deployed model identifier from the session interface. The following is a usage summary, not a transcript.

For Requirement 1, the user requested justification of cooling-efficiency quantities and their transfer from food storage to public cooling. Codex organized energy, service quantity, quality and duration, reviewed sources and drafted the explanation. A user-supplied Google Gemini analysis informed the process-oriented organization; its exact model identifier was unavailable.

At the user's request, the 2025 Problem B papers of Teams 16442, 16738 and 16739 were read as structural examples. Their estimates, weighting methods and wording were not adopted. The user also supplied freezer, shelter and Canberra schematics. Codex incorporated them as separate PDFs, checked available dimensional sources and distinguished architectural evidence from modeled occupied area and operating assumptions.

For Requirement 2(a), Codex implemented a three-zone bus scenario with psychrometric calculations and published SET, generated charts and CSV outputs, and developed sensitivity and accounting checks. For Requirement 2(b), Codex implemented dynamic background heat/moisture and surface balances, variable counts, allocation and coverage, comparing candidates against a matched no-device baseline.

After the user supplied Cruz-Salas et al.'s research and requested clearer model inheritance, Codex developed a shared local delivery operator, connected it to background balances, added heat-allocation and airflow sensitivities, tested conservation and limiting cases, reran comparisons, and revised figures and prose. AI review checked equations, units, implementation and figure consistency. The cafeteria study's accuracy was not transferred to Canberra.

All numerical outputs are conditional simulations. No field measurements, real installation inventory or experimental validation were produced; independent student verification has not been asserted. Geometry, airflow, mixing, coverage and occupancy assumptions remain identified in the paper and scenario files.

Before submission, add exact prompts and complete outputs for retained AI contributions, including Gemini, and complete tool/model details required by the contest guidance [22]. Retain this source and conversation as records. Requirements 3–5 and the final summary remain unfinished.